Preprint

SUPPLEMENTARY MATERIAL

G's Graded Operator Is a Null Rotation

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Abstract
Gravity Field Dynamics writes Newton's constant as $G = \frac{17}{13}\,\alpha^{21}\,\hbar c/m_e^2$, read from a graded operator on blocks of size 1, 4 and 16 whose traces are the coupling polynomial $f(K,s) = 1 + sK + K^2$. The archive's diagram draws those blocks as Sierpiński triangles. We show what they are. Each block is $S_m = A^{\otimes m}$, $A = \left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$, on $m = 0, 2, 4$ bits, equal to e^L with L the operator that removes one bit. G is unchanged by them and reads only their diagonal: $1 + 4 + 16 = 21$, while the filled area $1 + 9 + 81$ would miss G by 10^{-150}. Reading the two bits of the 4-block as the spinor indices of a 4-vector, the Sierpiński block is a null rotation, a Lorentz transformation that fixes one light ray. The Walsh–Hadamard read on the same block is a rotation of space by π, and it factors as null rotation, boost of rapidity $\ln 2$ with a π twist, and reverse null rotation. The trace of a null rotation on a 4-vector is $|\mathrm{Tr}\,A|^2 = 4 = K$, the boundary between rotations and boosts, so G's graded trace is the character $1 + s\chi + \chi^2$ of a null rotation on scalar, vector and rank-2 tensor. The Sierpiński chains split the blocks as $4 = 3 + 1$ and $16 = (5 + 3 + 1 + 1) + (3 + 3)$, the pieces of a metric and of a 2-form. Finally, the archive's tetrad frame, a row of ones over the four TetA corners, is the same Walsh–Hadamard read on the 4-block, so it factors through the null rotation. Read with its two bits as spinor indices, its four legs are a Newman–Penrose null tetrad: two real light rays and a complex-conjugate null pair. No spinor reading makes more than two legs real light rays.

Keywords: gravitational constant, Sierpiński triangle, Walsh–Hadamard transform, null rotation, Lorentz group, coupling polynomial, proof

Setting

The archive derives G from the graded operator $I_s = P_0 + sP_1 + P_2$ on blocks of size 1, $K = 4$ and $K^2 = 16$, the scalar, vector and tensor parts of the tetrad equation [1]. Its traces are $f(K,s) = 21, 17, 13$ at $s = +1, 0, -1$ (TetA, the Field Origin, TetB), and $\det(\alpha^{I_s}) = \alpha^{\mathrm{Tr}\,I_s}$ gives $G = \frac{17}{13}\,\alpha^{21}\,\hbar c/m_e^2$. The archive's diagram of this operator [2] draws the three blocks as Sierpiński triangles, and its account of the cube says that the blocks "are" the Sylvester–Hadamard matrices H_1, H_4, H_{16} [3]. We take the diagram's blocks at face value and determine what they are. Write $A = \left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$, $H_2 = \left(\begin{smallmatrix}1&1\\1&-1\end{smallmatrix}\right)$, $S_m = A^{\otimes m}$, and let $|x|$ be the number of 1-bits of an index x.

The Blocks

Theorem 1 (the Sierpiński blocks). *The diagram's blocks are S_0, S_2 and S_4. Entry (i,j) of S_m is 1 exactly when the bits of j are a subset of those of i, that is, when $\binom{i}{j}$ is odd. With $L = \sum_b N_b$, where N_b removes bit b (it acts as $\left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right)$ on that bit),*

$$S_m = e^L, \qquad L^{m+1} = 0, \qquad \mathrm{Tr}\,L = 0. \tag{1}$$

The traces are $1, 4, 16$ and the filled areas are $1, 9, 81 = 3^m$.

Proof. The diagram's cells are compared with S_m one by one. $A = e^N$ with $N^2 = 0$, and the N_b commute, so $A^{\otimes m} = \prod_b e^{N_b} = e^L$. The entry rule is the tensor product of A's and Lucas's theorem. Each bit contributes 1 to the trace and 3 to the area. □

Theorem 2 (G reads the diagonal). *Let $M_s = S_0 \oplus s\,S_2 \oplus S_4$. Then $\mathrm{Tr}\,M_s = 1 + 4s + 16 = f(K,s)$, the spectra of M_s and I_s coincide (rank 17 at $s = 0$; 17 eigenvalues $+1$ and 4 eigenvalues -1 at $s = -1$), and $\det(\alpha^{M_s}) = \alpha^{f(K,s)}$. So G is the same whether the blocks are identities or Sierpiński blocks. Reading the filled cells instead, $1 + 9 + 81 = 91$, would change G by $\alpha^{70} \approx 10^{-150}$.*

Proof. Each S_m is lower triangular with unit diagonal, so its eigenvalues are all 1. $\det e^X = e^{\mathrm{Tr}\,X}$. □

So the Sierpiński structure off the diagonal is invisible to G. The next theorems say what that structure is.

The Lorentz Reading

Write a 4-vector as $X = t + x\sigma_1 + y\sigma_2 + z\sigma_3$, so that $\det X = t^2 - x^2 - y^2 - z^2$, and index its entries by the two bits of the 4-block, $X = (x_{ij})$ [5]. The states $|00\rangle$ and $|11\rangle$ are $t + z$ and $t - z$, the two null directions along z.

Theorem 3 (the Sierpiński block is a null rotation). *On the bit basis, $S_2\,\mathrm{vec}(X) = \mathrm{vec}(AXA^T)$. The map*

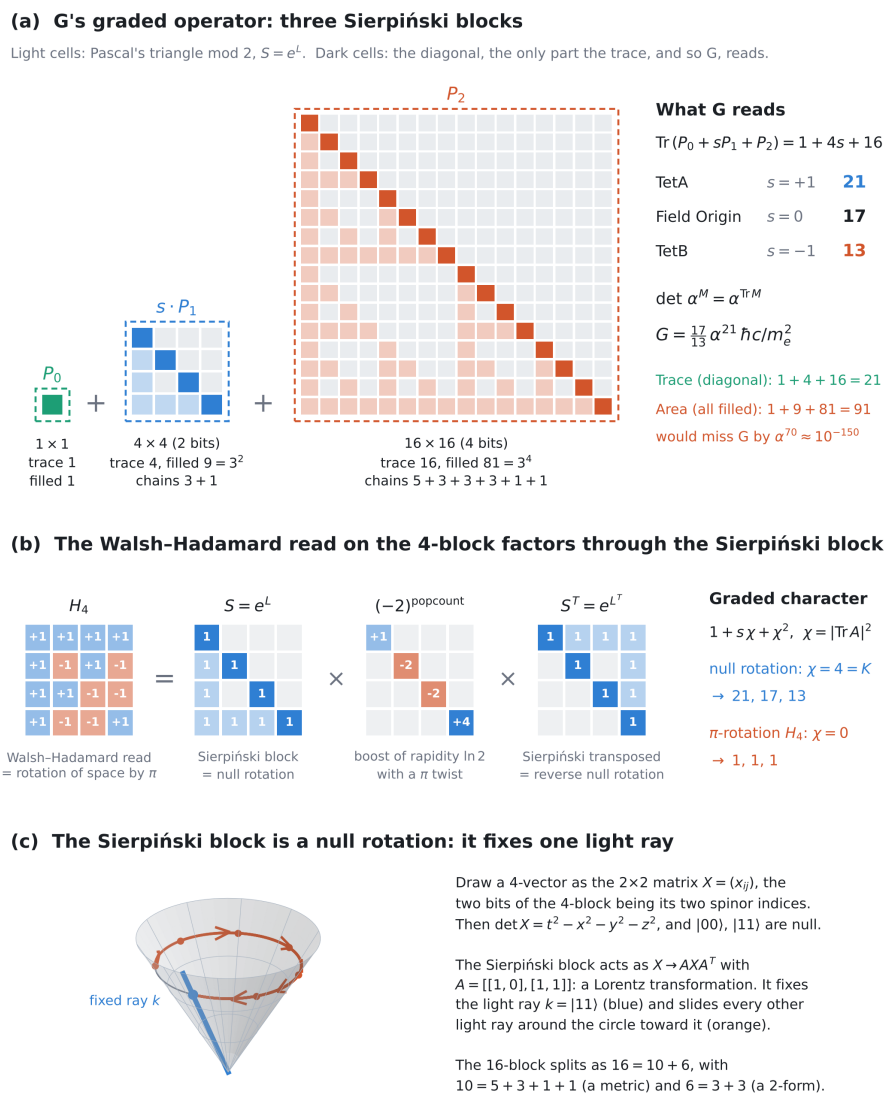


Figure 1. (a) G's graded operator as drawn in the archive: the Sierpiński blocks S_0, S_2, S_4. Their diagonals (dark) are all that the trace, and so G, reads. (b) The Walsh–Hadamard read on the 4-block factors through the Sierpiński block (Theorem 4). (c) Read as acting on 4-vectors, the Sierpiński block fixes the light ray $|11\rangle$ and carries every other light ray towards it (Theorem 3), shown on the slice $y = 0$, which it preserves.

$\Lambda_A : X \mapsto AXA^\dagger$ is a proper orthochronous Lorentz transformation with $\Lambda_A \neq 1$ and $(\Lambda_A - 1)^3 = 0$: a null rotation. Its fixed vectors form the plane spanned by $k = (1, 0, 0, -1)$, the state $|11\rangle$, and the y axis, and k is the only light ray in that plane. Every real A with $\det A = 1$ fixes y, so the Sierpiński blocks act as the Lorentz group of (t, x, z).

Proof. $\mathrm{vec}(AXA^T) = (A \otimes A)\,\mathrm{vec}(X)$, and A is real, so $A^T = A^\dagger$. $\det(AXA^\dagger) = \det X$ because $\det A = 1$. A fixes $|1\rangle$, so Λ_A fixes the matrix $|1\rangle\langle 1|$, which is the light ray k. The antisymmetric part $y\,i\sigma_2$ is sent to $\det A \cdot y\,i\sigma_2$. The remaining statements are computed from the 4×4 matrix of Λ_A. □

Theorem 4 (the Hadamard read). *The normalised $H_2/\sqrt{2}$ acts on 4-vectors as a rotation of space by π about $(\hat{x} + \hat{z})/\sqrt{2}$. For $m = 0, 2, 4$,*

$$H_{2^m} = S_m \,\mathrm{diag}\big((-2)^{|x|}\big)\, S_m^T = e^L\, e^{(\ln 2 + i\pi)|x|}\, e^{L^T}. \tag{2}$$

On the 4-block the middle factor $\mathrm{diag}(1, -2)/\sqrt{2}$ is a boost of rapidity $\ln 2$ along z combined with a rotation by π about z. So the rotation by π is a null rotation, a boost with a π twist, and the reverse null rotation.

Proof. For one bit, $A\,\mathrm{diag}(1, -2)\,A^T = H_2$; tensor powers give the rest, and $(-2)^{|x|} = e^{(\ln 2 + i\pi)|x|}$. The Lorentz matrices follow from $\Lambda_M = \frac{1}{2}\mathrm{Tr}(\sigma_\mu M \sigma_\nu M^\dagger)$, with

$\cosh \ln 2 = \frac{5}{4}$ and $\sinh \ln 2 = \frac{3}{4}$. □

This is the triangular (Gauss) factorisation of the Walsh–Hadamard matrix [7]. In it, the Sierpiński blocks are the outer factors and the half-turn $e^{i\pi}$ sits in the middle.

The Character

Theorem 5 ($K = 4$ is the trace of a null rotation). *For $M \in SL(2,\mathbb{C})$ the trace of Λ_M on 4-vectors is $\chi = |\operatorname{Tr} M|^2$, and on rank-2 tensors it is χ^2. So the graded trace of M acting on scalar, vector and tensor, with the sign s on the vector, is*

$$1 + s\,\chi + \chi^2 = f(\chi, s). \tag{3}$$

The null rotation has $\chi = 4 = K$, giving $21, 17, 13$. The π-rotation $H_2/\sqrt{2}$ has $\chi = 0$, giving 1. A rotation by ϕ has $\chi = 4\cos^2(\phi/2) \le 4$ and a boost of rapidity η has $\chi = 4\cosh^2(\eta/2) \ge 4$, with equality only for the identity: the null rotations lie on the boundary $\chi = 4$.

Proof. Λ_M is $M \otimes \bar{M}$ in the bit basis, so its trace is $\operatorname{Tr} M \overline{\operatorname{Tr} M}$, and the trace of a tensor product is the product of traces. $\operatorname{Tr} A = 2$, $\operatorname{Tr} H_2 = 0$, and the rotation and the boost about z are $\operatorname{diag}(e^{\pm i\phi/2})$ and $\operatorname{diag}(e^{\pm\eta/2})$. □

This gives G's three traces a reading as a character. It does not derive $K = 4$, which comes from $D = 3$ [4], and the identity has the same character. What it shows is that the blocks the archive draws, with their Sierpiński structure, are null rotations, and that the Walsh–Hadamard read itself, with $\chi = 0$, could not give G's numbers.

Theorem 6 (the chains). *The Jordan chains of S_2 have lengths $3 + 1$, and those of S_4 have lengths $5 + 3 + 3 + 3 + 1 + 1$. They are the Clebsch–Gordan decompositions of 2 and 4 spin-$\frac{1}{2}$ factors, and over the whole operator $21 = 1 + (3 + 1) + (5 + 3 + 3 + 3 + 1 + 1)$. S_4 commutes with the exchange of the two vector factors, and on the symmetric 10 and antisymmetric 6 its chains are*

$$10 = 5 + 3 + 1 + 1, \qquad 6 = 3 + 3. \tag{4}$$

Proof. L is the lowering operator of the diagonal $\mathfrak{sl}_2$ on m spin-$\frac{1}{2}$ factors, so each irreducible summand of dimension d is one chain of length d. The multiplicity of dimension $m - 2k + 1$ is $\binom{m}{k} - \binom{m}{k-1}$. The restriction to the symmetric and antisymmetric parts is computed from the ranks of $(S_4 - 1)^j$. □

The split $16 = 10 + 6$ is the archive's: general relativity keeps the symmetric 10, and the antisymmetric 6 is the torsion-type part. The chains refine it into the pieces of a symmetric tensor $(5 + 3 + 1 + 1)$ and of a 2-form $(3 + 3)$.

The Tetrad Frame

Theorem 7 (the frame is the 4-block read). *The archive's tetrad frame [4], a row of ones over the four TetA corners $(1,1,1)$, $(1,-1,-1)$, $(-1,1,-1)$, $(-1,-1,1)$, is $H_2 \otimes H_2$ with its second and third rows exchanged. So it factors through the Sierpiński block,*

$$e = P\,S_2\,\operatorname{diag}\big((-2)^{|x|}\big)\,S_2^T, \tag{5}$$

with P the row exchange: a null rotation, a boost of rapidity $\ln 2$ with a π twist, and the reverse null rotation. Read in (t, x, y, z), the frame is not a Lorentz structure: $e^T \eta\, e = 2J - 4I$, which is J minus the archive's forced Gram matrix $4I - J$.

Proof. Direct comparison, and Theorem 4 with $m = 2$. □

So the same 4×4 matrix plays two roles. As an operator on the 4-block it is a Lorentz transformation, the rotation by π. As a set of four legs it is the frame. The next theorem reads the legs in the spinor picture.

Theorem 8 (a Newman–Penrose null tetrad). *Read with its two bits as spinor indices, leg (a, b) of $H_2 \otimes H_2$ is the matrix $X = h_a h_b^T$, with h_0, h_1 the columns of H_2, so every leg has $\det X = 0$. In Minkowski components*

$$\begin{aligned} l &= (1,1,0,0), \qquad n = (1,-1,0,0), \\ m &= (0,0,-i,1), \quad \bar{m} = (0,0,i,1), \end{aligned} \tag{6}$$

with $l \cdot n = 2$, $m \cdot \bar{m} = -2$ and every other product 0: a Newman–Penrose null tetrad [6], built from the spinor dyad (h_0, h_1). Its Hadamard combinations $l \pm n$, $m + \bar{m}$ and $i(m - \bar{m})$ are $2t$, $2x$, $2z$ and $2y$. In any spinor reading, $X \mapsto BXC^T$ with B, C invertible, at most two of the four legs are real light rays.

Proof. The column (a, b) of $H_2 \otimes H_2$ is $h_a \otimes h_b = \operatorname{vec}(h_a h_b^T)$, a matrix of rank one. The components and products follow from $x^\mu = \frac{1}{2}\operatorname{Tr}(\sigma_\mu X)$ and the polarisation of det. For the bound: a rank-one matrix pq^T is Hermitian exactly when q is a real multiple of $\bar{p}$. If the legs $(a, 0)$ and $(a, 1)$ were both Hermitian, then Ch_0 and Ch_1 would both be multiples of $\overline{Bh_a}$, which is impossible for invertible C. So each a contributes at most one real light ray. □

So the frame's four legs are null in the spinor picture, in the form general relativity uses for

null tetrads: two real light rays and one complex-conjugate pair. This is a different reading from Letter 1's. There the legs are $(1, v/\sqrt{3})$, the Hadamard frame with its spatial rows rescaled by $1/\sqrt{3}$, and all four are real light rays on the celestial sphere. Both readings make the frame null; they differ in which coordinates carry the two bits.

Scope

The results are exact. They identify the operator from which G is read; they do not derive G. The value of G still rests on the archive's assumption that each of the 21 coupling levels contributes one factor of α [1], and on $K = 4$ from $D = 3$. The trace cannot tell a null rotation from the identity, so the content added here is what the blocks are: Sierpiński triangles, null rotations, and the outer factors of the Walsh–Hadamard read. The statement that the blocks "are" H_1, H_4, H_{16} is sharpened accordingly: they are the Sierpiński factors of those matrices, and the matrices themselves have trace 0 beyond H_1. The Sierpiński blocks act through the Lorentz group of (t, x, z), with y fixed. So the chain lengths match the decomposition of a vector, a metric and a 2-form exactly, but naming the components (h_{00}, h_{0i}, ...) requires the rotation group, which the Sierpiński block does not supply. The spinor–vector correspondence, null rotations and the factorisation of the Hadamard matrix are standard, as is the construction of a null tetrad from a spinor dyad; the identification with G's blocks and with the archive's tetrad frame is new. Theorem 8 gives two real null legs, not four, in the spinor reading. The Hadamard frame read directly in (t, x, y, z) has legs of norm -2; rescaling its spatial rows by $1/\sqrt{3}$, as Letter 1 does, makes all four legs real light rays.

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